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# Below the Masking Threshold: Observable Command-and-Control Discrimination in a Residual IRGC Missile Force

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Version 0.3, August 13, 2026. Table I H1/H2/H3 D_win levels bound to `data_phase2/phase2_discrimination.csv` (200-seed). Companion to the v4 attrition-masking paper [1].

## Abstract

A prior agent-based study found that under high launcher attrition, the daily ballistic-missile launch rate cannot distinguish an active command-and-control (C2) architecture from a pre-programmed one; only the emergent target ratio discriminates, and it is unobservable from open sources. This paper tests both sides of the boundary that finding implies, using an observable proxy: attribution correlation, whether Iran strikes the specific coalition asset that just provoked it, within a fixed window, and says so.

Three legs make the argument. Leg 2 codes the April-to-June 2026 ceasefire skirmish period under a pre-registered protocol and finds a Stream B attribution-correlation rate near 0.64 against a fixed-target control near 0.0, on an event count too small for an inferential test. Leg 1 extends the canonical model into a residual-force regime with a provocation-response mechanic and runs 200 Monte Carlo seeds; launch-rate discrimination stays null while attribution correlation separates the architectures sharply (rank-biserial r from 0.83 to 1.00, p less than 0.001). Leg 3 tests the discriminator after the ceasefire collapsed and nightly fighting resumed in July 2026. A pre-registered null simulation predicted that a pre-programmed force scores 0.55 to 0.72 under dense provocation; the observed rate of 0.500 falls below that band.

The combined result is a bounded operating envelope. Attrition masks targeting from above; provocation density saturates the proxy from below. Attribution correlation is a real but situational discriminator, not a standing indicator.

## I. Introduction

The 2026 US-Israeli campaign against Iran cut Iranian ballistic-missile launch rates by about 92 percent in nine days. A prior agent-based model (ABM) asked whether that fall reflected physical destruction of transporter-erector-launcher (TEL) assets or degradation of command authority, and concluded that launcher attrition explained the dominant share of the launch-rate variance [1]. The same work established a boundary. When week-one attrition is high, three competing C2 hypotheses (active distributed command, pre-programmed autonomous execution, and a mixed state) produce launch-rate curves that are statistically indistinguishable. The only reliable discriminator was the emergent target ratio, the fraction of strikes aimed at targets that became relevant only after the war began. That ratio needs pre-war target packages to compute, so it is a collection requirement, not an open-source measurement.

The boundary has two sides. The v4 paper characterized the high-attrition side, where targeting information is present in the surviving cells but the cells are destroyed before they can act on it. The low-attrition side was named but not tested: once a residual force operates past the acute-attrition phase, does its target selection become observably adaptive?

The April to June 2026 ceasefire period is that regime. Independent US intelligence reporting placed launcher survival at 40 to 50 percent at the April 8 ceasefire, roughly double the model's original assumption [1], and Iranian activity continued at low tempo through the following months. Attrition was no longer fast enough to swamp behavior. If target selection carries a C2 signature, this is where it should surface.

The signature we test is not the emergent target ratio, which stays unobservable, but a weaker observable proxy for it: attribution correlation. An event is attribution-correlated when Iran strikes the specific coalition asset that conducted an identifiable action against it within a fixed window, and Iranian framing links the strike to that action. Attribution correlation captures reactive retargeting. It is not the full adaptive-targeting construct, and the substitution is stated as a limitation, not smuggled in as equivalence.

Three legs make the argument. The empirical leg (Leg 2) codes the skirmish period under a pre-registered protocol and reports the attribution-correlation rate. The event count is small by construction, so it carries no inferential test. The simulation leg (Leg 1) extends the canonical ABM into the residual-force regime, adds a provocation-response mechanic, and runs enough Monte Carlo seeds to carry the statistics that the empirical record cannot. The two are compared qualitatively. A third leg (Leg 3), pre-registered before any July data was coded (DOI: https://doi.org/10.5281/zenodo.21443990), tests what happens when the discriminator's operating conditions break: the June 17 ceasefire collapsed, fighting resumed on July 7, 2026, and dense provocation saturated the construct (Section VI).

# II. The Prior Result

The v4 model represents 120 launch cells across five regions, each with a missile inventory, a pre-authorized target list, a pre-authorization expiry day, and a connectivity state to surviving C2 nodes [1]. Three behavioral rulesets encode the hypotheses. H1 (active distributed C2): connected cells receive updated targeting including conflict-emergent targets and prioritize the highest-value available target; disconnected cells fall back to pre-authorized packages. H2 (pre-programmed execution): all cells fire pre-authorized packages regardless of connectivity, and produce no emergent targeting. H3 (mixed): connected cells follow H1 with probability 0.65, otherwise H2.

Two findings from that work set up this one. First, across all phases and five pre-authorization windows, launch-rate differences between H1 and H2 were non-significant (Mann-Whitney U, p greater than 0.05; rank-biserial r less than 0.08). Launch rate is blind to C2 state under high attrition. Second, the emergent target ratio discriminated strongly (r greater than 0.60, p less than 0.001), and a post-conflict re-simulation under realistic 40 to 50 percent survival found the discriminator grew stronger in the residual-force regime, not weaker (r rising from 0.356 to 0.706) [1]. The re-simulation did not, however, connect the emergent ratio to anything an analyst could observe. That gap is what this paper addresses.

# III. Empirical Leg: Coding the Skirmish Period

The empirical leg is a pre-registered coding study of Iranian and Iran-attributed strikes from April to late June 2026. The unit is a retaliatory strike event: one discrete offensive action against a coalition, Gulf-state, or opposition target, bounded in time and target. A multi-target salvo is one event per distinct named target. Purely defensive maritime actions that enforce the strait blockade are logged separately and excluded from the construct.

Events are stratified into two streams before any construct is coded. Stream B, the primary stream, holds coalition and Gulf-state targets: US bases, US naval vessels, and Gulf-state territory and infrastructure. Coalition actions there create new, dated, attributable provocations, so adaptive targeting is testable. Stream A, the control, holds Iranian strikes on Kurdish opposition camps in Iraq, a target set Iran has struck for years. Stream A is a fixed-target baseline expected to show near-zero attribution correlation by construction, and it anchors the low end of the scale.

The construct is attribution correlation, coded 0, 0.5, or 1. A 1 requires both a physical match (the target is the specific asset that acted against Iran within 72 hours) and attributable Iranian framing (a statement from the IRGC, Iranian state media, or a named official linking the strike to that action). A physical match without framing, or framing without a dated antecedent, is 0.5. A standing target with no dated antecedent is 0. The 72-hour window is fixed and not re-tuned; a 48-hour robustness value is declared. Ambiguity defaults down.

Coding the skirmish period gives a Stream B attribution-correlation rate near 0.64 across seven construct events, against a Stream A control near 0.0. The Stream B signal rests on three clean events: the May 7 destroyer exchange, the May 30 Fateh-110 strike on Ali Al Salem Air Base after that base conducted the prior US strike, and the June 28 Kuwait and Bahrain salvo after US strikes on Iranian sites. Response latency across the three is short, on the order of a day. The event count is small, so no inferential test is claimed on it. The honest reading is that a consistent pattern of attribution-correlated retaliation exists in the low-attrition regime; three events cannot establish that the regime is adaptive. Supplying the statistics is the job of the simulation leg.

A retro-audit of provocation density around the three clean events sharpens the evidential hierarchy. B6 (May 30, Ali Al Salem) is the strongest: its 72-hour window contains one coalition action after a 23-day quiet gap, so the physical match (Iran named "the base responsible for the previous day's attack") carries real information against a low coincidence floor. B5 (May 7) is structurally different: the target and the provocation are the same object (the transiting destroyer force), so the coding captures a direct engagement rather than a retargeting decision. B9 (June 28) is the most exposed: its 72-hour window contains three coalition actions in a tit-for-tat cluster (June 25, 26, 27), so the existence of a temporal antecedent is certain by construction. The coding rests on Iran's specific framing and the physical match to Ali Al Salem, not the temporal coincidence alone. The sparse-provocation condition that gives B6 its weight was already weakening by late June, which is part of what motivates the Leg 3 out-of-sample test.

# IV. Simulation Leg: Design

The simulation extends the canonical model (`c2_core.py`) into the residual-force regime. The extension is additive: a bit-for-bit parity test against the original module reproduces all v1 and v3 behavior across 192 configurations, so nothing in the prior pipeline moves.

**Residual attrition.** A new attrition profile starts at about 42 percent launcher survival and drifts down slowly to about 38 percent over 60 days, holding the force past the acute-attrition phase. About 50 of the 120 cells survive.

**Provocation-response mechanic.** A provocation is a discrete timed event with a source asset and a response window. Half the provocations are new post-initiation assets, which a pre-programmed force has no package for and cannot strike. Half are standing pre-war targets, which a pre-programmed force can strike by coincidence through its normal firing. The standing half gives H2 a realistic non-zero coincidence floor and keeps the comparison from being decided by construction. When an adaptive cell (H1 or H3) is connected and a provocation window is open, it retaliates directly against the provoker with probability 0.65. H3 retaliates at that probability scaled by its 0.65 mixed-adaptation fraction. H2 never retaliates; it fires its standing list.

**Exogenous tempo.** Fire rate is held flat and equal across hypotheses. Every cell fires at the same low standing rate regardless of C2 state, and a fired shot always produces a launch. C2 state changes what is struck, never how much. This mirrors the v4 modeling choice and makes launch volume unable to discriminate the hypotheses by construction, so any launch-rate separation would be an artifact and its absence is a check, not an assumption.

**In-simulation metric.** Attribution correlation is the fraction of launches that strike a provocation source inside its window. We report two denominators. D_all divides by all launches over the 60 days, matching the empirical whole-period construction. D_win divides by launches fired while a provocation window is open, the pure response rate. D_win is the direct analog of the empirical construct (of the strikes fired during a provocation window, how many hit the provoker), so it is the primary reported metric; D_all is the conservative whole-period figure. The window is 3 days (72 hours) primary and 2 days (48 hours) for robustness. Response latency is recorded per correlated strike.

**Connectivity sweep.** The residual force is assumed to have re-established a flat, reconstituted C2 connectivity. That assumption carries weight, so it is swept rather than fixed, across connectivity levels of 0.20, 0.35, 0.50, and 0.65. Reporting discrimination across the sweep shows how far the result depends on the reconstitution assumption.

Each condition runs 200 Monte Carlo seeds. The provocation schedule is shared across hypotheses for a given seed and window, so the three rulesets see the same provocations and differ only in response. Discrimination between H1 and H2 uses the Mann-Whitney U test with a tie-corrected normal approximation, rank-biserial r as the primary effect size, and Cohen's d reported alongside. The stack is Python standard library only, and the grid runs in about 30 seconds.

## V. Results

The result is a contrast on identical runs. Table I reports H1 versus H2 discrimination at the primary 72-hour window across the connectivity sweep, for launch rate and for attribution correlation.

TABLE I. H1 vs H2 DISCRIMINATION, PRIMARY WINDOW (72h), 200 SEEDS

| Connectivity | r (launch rate) | p | r (D_win) | p | H1 D_win | H2 floor | H3 D_win |
|---|---|---|---|---|---|---|---|
| 0.20 | 0.001 | 0.980 | 0.934 | < 0.001 | 0.174 | 0.034 | 0.113 |
| 0.35 | -0.054 | 0.348 | 0.994 | < 0.001 | 0.249 | 0.034 | 0.176 |
| 0.50 | -0.008 | 0.896 | 1.000 | < 0.001 | 0.348 | 0.034 | 0.240 |
| 0.65 | 0.036 | 0.533 | 1.000 | < 0.001 | 0.433 | 0.034 | 0.303 |

Launch-rate discrimination is null at every connectivity level (p greater than 0.05), reproducing the v4 launch-rate finding inside the residual regime. Attribution correlation separates the two architectures at every level (p less than 0.001; Cohen's d from about 2.5 at the lowest connectivity to about 5 at the highest). Fig. 1 (figure omitted in this preprint; see `figures/phase2_fig1_discrimination.png`) shows the contrast in panel (a) and the by-hypothesis signal in panel (b).

The connectivity sweep sets magnitude, not the qualitative result. H1 in-window attribution correlation rises from 0.17 to 0.433 as reconstituted connectivity rises from 0.20 to 0.65, and H3 tracks below it (0.11 to 0.30), while the H2 coincidence floor holds near 0.034. Even at the lowest connectivity, where an adaptive force is in contact only one time in five, the proxy still separates H1 from H2 (r equal to 0.934, p less than 0.001). The 48-hour robustness window reproduces the pattern: launch rate stays null, and attribution correlation stays significant at every level (r from 0.837 to 1.000, all p less than 0.001), with a shorter one-day median latency. Response latency in the simulation is one to two days across the sweep, matching the roughly one-day latency of the three clean empirical events.

# VI. Leg 3: Out-of-Sample Test Under Resumed Hostilities

The low-attrition window this paper studies was bounded. The June 17 Islamabad MOU, which formalized the April 8 ceasefire and reopened the Strait of Hormuz, collapsed after Iranian strikes on commercial shipping. Major combat operations resumed on July 7, 2026, with nightly US strike packages and multi-country Iranian retaliation. Leg 3 was pre-registered before any July data was coded.

## VI-A. The Dense-Provocation Problem

The Leg 2 rate of 0.64 was measured in a sparse-provocation regime where coalition actions were rare and discrete. The July regime broke that. The US struck Iran on 17 or more nights in the 31-day window (July 7 to August 6). The 72-hour antecedent window was permanently open. Any Iranian strike on any US-linked target had a dated antecedent within 72 hours, mechanically. IRGC framing was systematic, with numbered retaliatory phases ("Nasr 24" and subsequent operations) attached to nearly every salvo. Under these conditions the construct's null expectation is not near zero.

A Monte Carlo of the coding scheme (`leg3_null_sim.py`, 1,000 events per cell) asked what a purely pre-programmed force scores on this construct when provocations occur daily and framing is universal. The answer is 0.55 to 0.72, depending on the framing probability and the fraction of antecedent strikes attributable to a specific basing asset rather than a carrier or standoff platform. The mechanism is structural: framing alone earns 0.5 per event, and with daily provocations originating from the same basing set Iran strikes, chance physical matches supply the rest.

## VI-B. Results

Leg 3 coded 31 construct events in Stream B and 5 representative events in Stream A, using the codebook without modification.

**Stream B.** Six events coded 1 (Ali Al Salem and 5th Fleet on July 8, Azraq on July 9 and July 21, Jordan on July 18, and Muwaffaq Salti on July 30). Nineteen coded 0.5. Six coded 0. One non-military target (an Amazon data center in Bahrain) was excluded per author review. The raw attribution-correlation rate was 0.500, below the pre-registered null band of 0.55 to 0.72.

Fifteen of 31 events (48 percent) were ceiling-limited at 0.5 by the antecedent-unattributable flag: Iran could not identify or reach the carrier or standoff platforms that delivered the antecedent strikes, so the strict physical-match requirement was structurally unsatisfiable regardless of how adaptive the targeting was. This deflation pulls the observed rate below the null band that a pre-programmed force achieves by coincidence; the saturation inflation that the null simulation models is more than offset by the deflation the simulation does not model.

**Stream A.** All five events scored 0. Iran claimed none and framed none as retaliation, despite every event falling within 72 hours of at least one US strike on Iran. The July 17 missile strike on the Komala camp near Sulaymaniyah killed nine, the deadliest single Kurdish strike in the window, and went entirely unclaimed.

**Framing asymmetry.** This is the cleanest finding. In a window where every Iranian strike had a 72-hour US antecedent available, Iran framed every Gulf strike as retaliation and claimed zero Kurdish strikes as retaliation. The asymmetry is total and selective: attribution framing is performed for the Gulf audience and withheld for the Kurdish audience even when identical antecedent conditions hold. This bears on the attribution-laundering threat (codebook Section 8 item 2): the more systematically Iran performs retaliation framing, the harder it is to tell genuine adaptive targeting from propaganda overlaid on pre-planned execution.

**Robustness.** The signal is geographically concentrated on Jordan. Azraq Air Base, which hosts US strike aircraft flying Iran sorties, appears in four events (two coded 1). Removing all Azraq events drops the rate to 0.463. Removing all Jordan events (Azraq, Muwaffaq Salti, Prince Hassan, and the July 18 KIA event) drops the rate to 0.409. The concentration is consistent with adaptive selection of the most accessible US strike platforms, but it means the finding's base is narrow.

Latency across the six clean 1s: median 24 hours. Identical to Leg 2.

### VI-C. Interpretation

H_ceiling is the outcome. The raw attribution-correlation rate falls below the null band that a pre-programmed force achieves by coincidence under dense provocation, pulled down by the deflation effect of unattributable antecedents. The construct cannot discriminate C2 state in this regime: the rate is uninformative whether it lands inside or below the band, because both saturation and deflation are artifacts of the dense-provocation environment rather than signals of C2 architecture.

Combined with the Leg 1 and Leg 2 result, the picture closes. Under high attrition (the v4 paper), launch rate is blind to C2 architecture. Under low attrition and sparse provocation (Legs 1 and 2), an observable proxy for target selection separates the architectures cleanly. Under low attrition and dense provocation (Leg 3), the observable proxy saturates and loses discriminating power. OSINT C2 discrimination has a bounded operating envelope. Attrition masks targeting from above. Provocation density saturates the proxy from below. The sparse, low-attrition window of April to June 2026 fell inside that envelope. The construct's value is real but situational, and it cannot be relied on as a standing indicator.

## VII. Discussion

The finding is a boundary flip. Under high attrition, launch rate and target selection are both blind or unavailable, and C2 architecture is masked. Under the residual-force regime, launch rate is still blind, but an observable proxy for target selection separates the architectures cleanly. The discriminator that the v4 paper could name only as an unobservable collection requirement has an observable counterpart once attrition stops masking targeting.

**Comparison to the empirical record.** The simulation and the coding study measure the same construct and agree in direction. Iran, in the skirmish period, struck the specific provoker within a short window and framed it as retaliation, at a Stream B rate near 0.64 against a fixed-target control near 0.0. In simulation, an adaptive force produces exactly this pattern while a pre-programmed force cannot, and the response latency matches. The comparison is qualitative and bounded. The empirical 0.64 sits above the simulated H1 in-window rate (0.433 at the highest connectivity) because the empirical denominator counts only construct-coded events, a more selective population than every in-window launch in the simulation. The levels are not meant to coincide; the direction and the separability are the claim. The empirical Stream B, read against the simulation, is consistent with an adaptive-leaning residual force, not with pure pre-programmed execution.

**What would falsify this.** The simulation is falsifiable in the same spirit as its parent. If launch rate had separated the hypotheses, the exogenous-tempo construction would be wrong and the null check would have failed; it did not. If attribution correlation had collapsed toward the launch-rate null once the H2 coincidence floor was admitted, the proxy would carry no signal and the empirical 0.64 could not discriminate; it did not. The empirical leg carries its own falsification discipline: a low Stream B rate would have been reported as evidence against observable adaptation, not explained away.

**Limitations.** Three bound the result. First, attribution correlation is a proxy. It captures reactive retargeting, not the full adaptive-targeting construct; a force could be adaptive without striking the exact provoker, and a force could claim precise retaliation for propaganda even when targeting was pre-planned. The coding protocol requires both a physical antecedent match and attributable framing to limit the second failure, but the proxy is weaker than the emergent ratio it stands in for. Second, the near-complete separation at higher connectivity (r at or near 1.0) reflects idealized behavioral archetypes. H1 and H2 are pure types; a real force is a mixture, which is what H3 models and what the empirical 0.64 reflects. The simulation shows the discriminator works, not that real-world discrimination is perfect. Third, the reconstituted-connectivity assumption is swept, not measured. The result holds across the swept range, but the true residual connectivity of the IRGC in this period is not known from open sources.

**Collection implication.** Attribution correlation is observable where the emergent target ratio is not. It needs only dated coalition actions, dated Iranian strikes, and attributable Iranian framing, all of which appear in open reporting. That makes it a candidate indicator for the low-attrition, sparse-provocation regime, to be collected and coded during a campaign lull rather than reconstructed after declassification. It does not replace the emergent target ratio as the theoretically sound discriminator; it is the observable shadow the emergent ratio casts when a residual force retaliates against fresh provocations. Leg 3 sharpens the caveat: the indicator works only when provocations are sparse enough that an antecedent match carries information. Under nightly strikes the shadow blurs into noise.

**The framing asymmetry as a standalone finding.** Even where the attribution-correlation construct saturates, the framing asymmetry (Gulf strikes claimed as retaliation, Kurdish strikes unclaimed) persists and is itself informative. It does not discriminate C2 state directly, but it constrains inference about Iranian targeting doctrine: the IRGC treats the Gulf and Kurdish audiences as separate propaganda domains, performing attribution for one and withholding it from the other. This selective framing is evidence that retaliation claims serve a political function independent of actual targeting decisions, which is relevant to any future use of framing-based indicators.

**Munitions and sustainability.** By early August 2026, CNN reported that nearly 80 percent of the THAAD interceptor inventory and roughly half of Patriot stocks had been expended since February 28, with CSIS estimating 3 or more years to rebuild THAAD stockpiles. The v4 paper's conditional recommendation (prioritize left-of-launch attrition under uncontested ISR) assumed a sustainable interceptor magazine. That assumption has failed empirically. The binding constraint on coalition operations has shifted from ISR dominance to interceptor supply. This matters for the present paper because the sparse-provocation window that made attribution correlation useful (Legs 1 and 2) existed partly because interceptor scarcity forced the coalition to limit strike tempo. If interceptor stocks had held, the dense-provocation regime that saturated the construct (Leg 3) might have arrived earlier, closing the observable window sooner.

# VIII. Conclusion

The 2026 campaign left a residual Iranian missile force operating at low tempo and moderate attrition through the ceasefire period. In that regime, the launch rate remains blind to command-and-control state, as it was under high attrition, but target selection does not. A pre-registered coding of the skirmish period finds Iranian retaliation correlated with specific coalition provocations at a Stream B rate near 0.64 against a fixed-target control near 0.0, on an event count too small to test. A residual-force simulation, run at proper Monte Carlo scale on identical runs, reproduces the launch-rate null and shows that an observable attribution-correlation proxy separates an active from a pre-programmed architecture at every connectivity level tested, with a response latency that matches the empirical events.

When the ceasefire collapsed and nightly fighting resumed in July 2026, the discriminator saturated. A pre-registered null simulation predicted that a purely pre-programmed force would score 0.55 to 0.72 on the construct under daily provocation, and the observed Leg 3 rate of 0.500 fell below that band, pulled down by the deflation effect of unattributable standoff and carrier antecedents that the null simulation does not model. The coding scheme cannot tell adaptive from pre-programmed targeting when antecedents are always available and framing is universal. The framing asymmetry, Iran claiming all Gulf strikes as retaliation while leaving all Kurdish strikes unclaimed, is the residual signal, but it speaks to propaganda function rather than C2 architecture.

The combined result across three legs is a bounded operating envelope. Attrition masks C2 architecture from above (the v4 finding). Provocation density saturates the observable proxy from below (Leg 3). Between the two, in the sparse and quiet regime where provocations are rare enough that an antecedent match carries information, the discriminator works (Legs 1 and 2). That window was real: April to June 2026 fell inside it, and the signal there is consistent with adaptive targeting. But the window was temporary, and the construct cannot be relied on as a standing indicator. Any future use of attribution correlation as a C2 diagnostic requires first establishing that the provocation environment is sparse enough to stay below the saturation ceiling this paper quantifies.

# AI Utilization Statement

This work was produced with AI assistance. The author used Claude (Anthropic) for simulation code extension, statistical pipeline construction, figure generation, and manuscript drafting. All substantive claims, analytical decisions, and final editorial judgments were made by the author. AI-generated content was reviewed and corrected by the author before inclusion. No AI system is listed as a co-author.

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# Dedication

Dedicated to Prof. Eric Nordlinger a”h. Thank you for everything.

Eric A. Nordlinger died at his home in Cambridge, Mass., June 3, 1994. He was 54. Dr. Nordlinger earned his undergraduate degree from Cornell. He became a professor of political science and an associate at the Center for Foreign Policy Studies at Brown, where he had been a member of the faculty since 1972. He was also an associate of the Center for Intl. Affairs at Harvard.

# References

[1] D. Y. Bilar, "Launcher Attrition Dominates Command Architecture: Agent-Based Analysis of IRGC Missile C2 Degradation in the 2026 Iran War," Zenodo, 2026. doi: 10.5281/zenodo.19558036. [Online]. Available: https://doi.org/10.5281/zenodo.19558036

*Prerelease note: Empirical event sources for Sections III and VI (CENTCOM, ISW/CTP, JINSA, CNN, Iranian state-media framing statements, and Leg 3 sources including Al Jazeera, IranWire, Xinhua, JNS, FDD, KHRN, and GlobalSecurity) are documented per event in the companion coding tables and codebook. They require primary-source verification and IEEE formatting before submission and are omitted here rather than listed unverified.*